

Electrochemical Gradients Assessment Device (EGAD)

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Action Potentials Extended Learning Objectives Assessed

Learning Objective	Assessment item
1. Explain that diffusion is the net movement of particles from a region of high concentration to a region of low concentration based on the random motion particles.	Q1/Q2
2. Predict and explain ion movement in response to diffusion and electric forces including application of the concept of the equilibrium potential (potential at which diffusion and electric forces are in balanced opposition, resulting in no net flow of ions).	Q3/Q4, Q5/Q6, Q7/Q8, Q9/Q10
3. Predict and explain changes in membrane potential based on membrane conductance to Na^+ and K^+ and ion movement.	Q9/Q10, Q13, Q14
4. Predict and explain changes in membrane potential based on concentrations of Na^+ and K^+ and ion movement.	Q12/Q13
5. Explain that transmission between neurons usually uses chemical neurotransmitter.	Q15
6. Explain that action potential firing rate reflects stimuli intensity.	Q16
7. Explain that neurons have stimulus thresholds for producing action potentials.	Q17

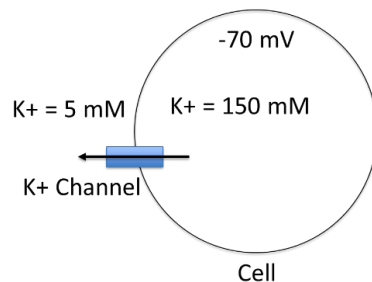
EGAD Assessment Questions

1. During the process of diffusion, particles will generally move from areas of
 - a. high to low concentration.
 - b. low to high concentration.
2. The reason for my answer is because
 - a. the particles are crowded and want to move to reach equilibrium.
 - b. the particles move randomly which over time results in their uniform distribution.
 - c. the particles tend to keep moving until they are uniformly distributed and then they stop moving.
 - d. the particles have a greater chance of bouncing off each other.

When the ratio of positively to negatively charged particles is different inside and outside of the cell membrane, we say there is a **membrane potential** (measured in mV). When charged particles, like ions, move into or out of a cell, the membrane potential changes.

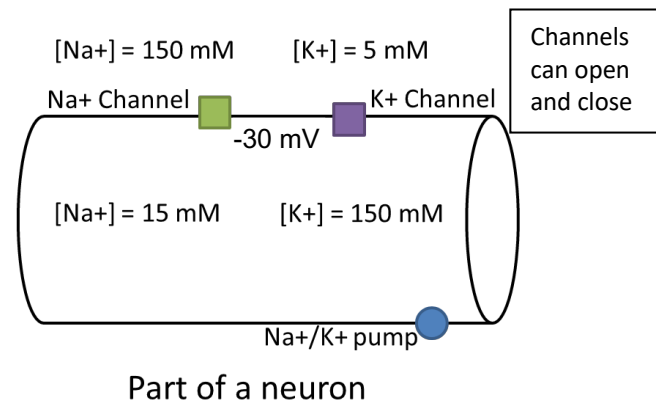
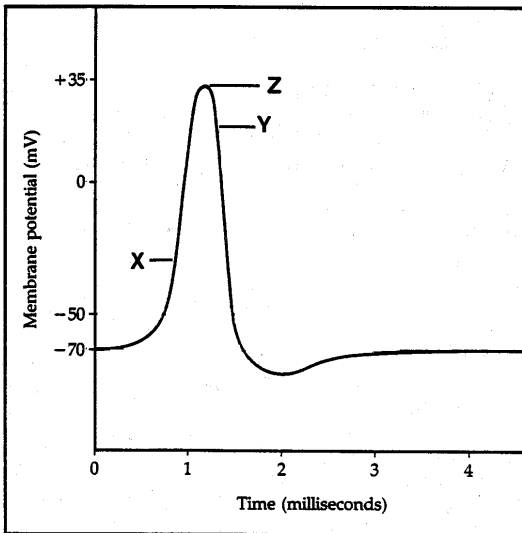
- When positive ions move into the cell, the membrane potential becomes more positive
- When positive ions move out of the cell, the membrane potential becomes more negative

Here is a diagram of a normal cell. The labels indicate the concentrations of K^+ ions inside and out of the cell, the resting membrane potential and K^+ channels. A channel is a protein that allows ions to move freely into and out of the cell.

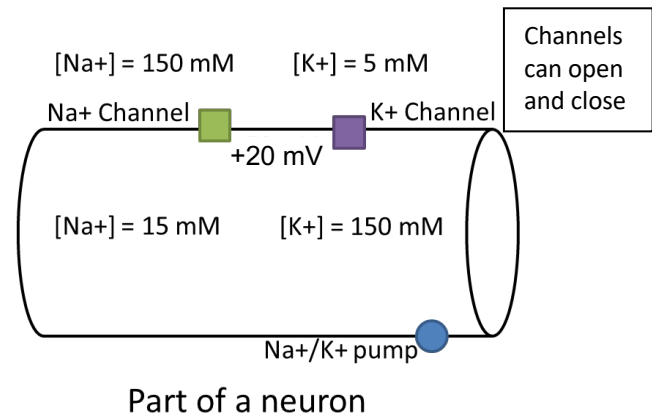
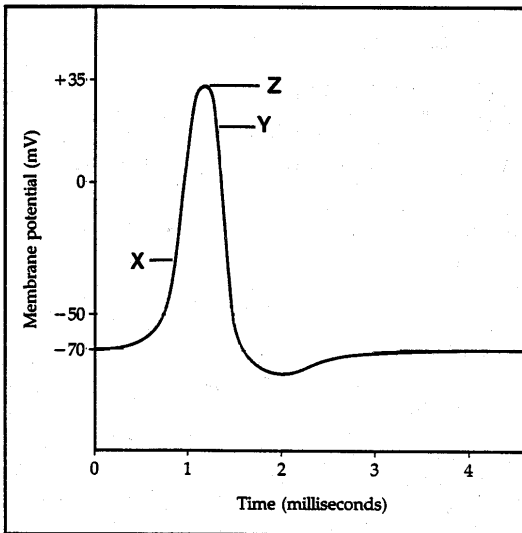


3. When the K^+ channels are open, there is net movement of K^+ ions out of the cell (as indicated by arrow). When will the net movement of K^+ out of the cell **STOP**?
 - a. When the concentration of K^+ inside and outside of the cell are the same.
 - b. When the membrane potential becomes more positive, for instance -50 mV .
 - c. When the membrane potential becomes more negative, for instance -90 mV .
4. The net movement of K^+ out of the cell stops because
 - a. the K^+ ions are no longer crowded inside the cell.
 - b. K^+ concentrations inside and outside of the cell have reached equilibrium.
 - c. the electric forces inside and outside of the cell have reached equilibrium.
 - d. the electric forces are equal and opposite to the K^+ concentration gradient.

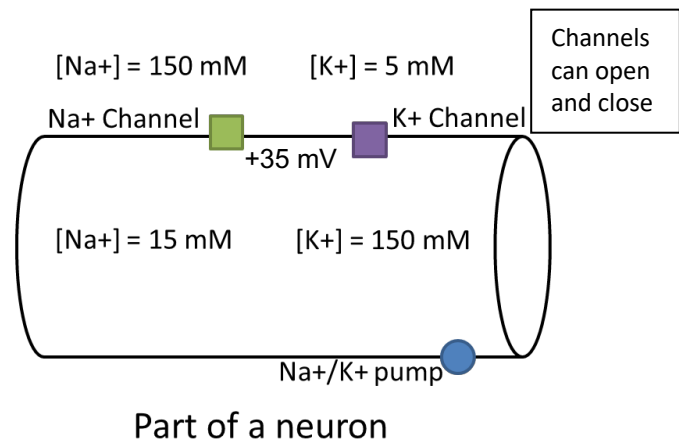
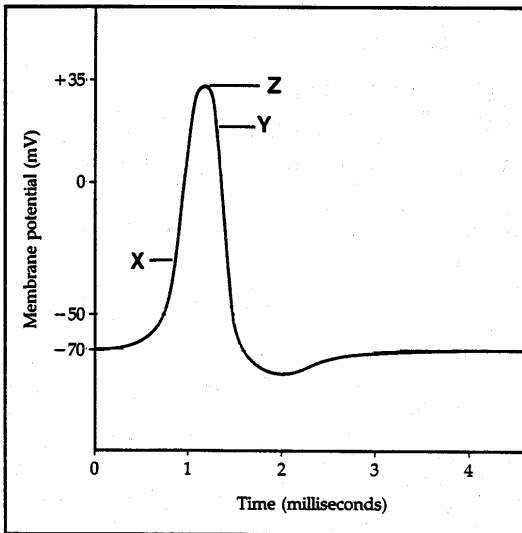
Action potentials are electrical signals on the cell membranes of neurons. They are generated by ions moving across the membrane of the neuron. Below is a figure showing changes in membrane potential during an action potential. There is also a diagram of a part of a neuron with labels indicating Na^+ and K^+ ion concentrations, membrane potential, Na^+ and K^+ channels, and the Na^+/K^+ pump.



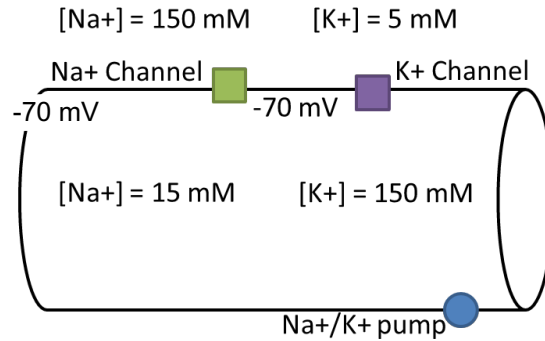
5. What causes the membrane potential to be more positive than -70 mV at the point labeled X?
 - a. Na^+ and K^+ channels are open and Na^+ and K^+ ions are rushing into the neuron.
 - b. Na^+ channels are open and Na^+ ions are rushing into the neuron.
 - c. K^+ channels are open and K^+ ions are rushing into the neuron.
6. Those ions are moving into the neuron because
 - a. by coming in they'll cause the membrane potential to become more positive.
 - b. the neuron needs to generate a more positive membrane potential.
 - c. of electric forces.
 - d. of the concentration gradient.
 - e. of the concentration gradient and the neuron needs to generate a more positive membrane potential.
 - f. of electric forces and the concentration gradient.



7. What causes the membrane potential to be more negative than +35 mV at the point labeled Y?
- Na^+ and K^+ channels are open and Na^+ and K^+ ions are rushing out of the neuron.
 - Na^+ channels are open and Na^+ ions are rushing out of the neuron.
 - K^+ channels are open and K^+ ions are rushing out of the neuron.
8. Those ions are moving out of the neuron because
- by moving out they'll cause the membrane potential to become more negative.
 - the neuron wants to return to a negative membrane potential.
 - of electric forces.
 - of the concentration gradient.
 - of the concentration gradient and the neuron wants to return to a negative membrane potential.
 - of electric forces and the concentration gradient.

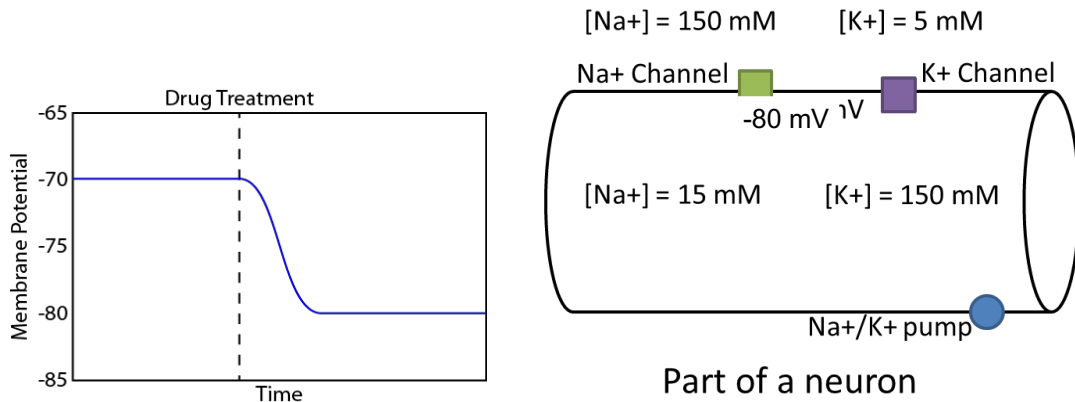


9. You add a drug to a neuron that keeps the Na⁺ channels open and the K⁺ channels closed at point Z. What happens to the height of this action potential?
- It will stay at +35 mV.
 - It will go up slightly (maybe +65 mV).
 - It will go up extremely high (maybe +300 mV or higher).
 - It will drop below + 35 mV.
10. The reason for my answer is
- Na⁺ ions will keep entering the neuron until the Na⁺ concentrations are at equilibrium.
 - Na⁺ ions will want to enter the cell down the concentration gradient and make the cell more positive.
 - Na⁺ ions will keep entering the neuron until the electric forces are equal and opposite to the Na⁺ concentration gradient.
 - the cell membrane potential cannot get more positive than +35 mV.
 - the cell membrane potential must eventually drop below +35 mV.



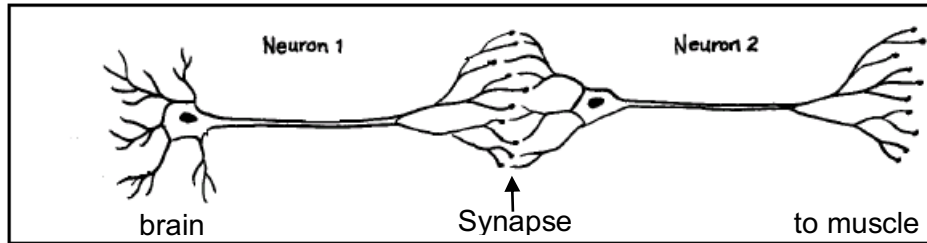
Part of a neuron

11. The resting membrane of a neuron has a few open Na^+ and K^+ channels and a membrane potential of -70 mV . What would happen to the resting membrane potential if the concentration of K^+ outside of the cell was increased to 50 mM ?
- It will be more positive than -70 mV .
 - It will be more negative than -70 mV .
 - It will stay the same at -70 mV .
 - It will be 0 mV .
12. The reason for my answer is
- more K^+ ions are now entering the cell.
 - fewer K^+ ions are now leaving the cell.
 - the Na^+/K^+ pump would keep the membrane potential at -70 mV .
 - there are now more K^+ ions outside of the cell.



You add a drug to a neuron (at the dashed line) and it causes the membrane potential to change, as seen in the graph.

13. What do you think the drug could be doing to the K⁺ permeability of the membrane?
 - a. Increasing K⁺ permeability of the membrane.
 - b. Decreasing K⁺ permeability of the membrane.
 - c. No impact on K⁺ permeability of the membrane.
14. What do you think the drug could be doing to the Na⁺ permeability of the membrane?
 - a. Increasing Na⁺ permeability of the membrane.
 - b. Decreasing Na⁺ permeability of the membrane.
 - c. No impact on Na⁺ permeability of the membrane.



15. There are two neurons between your brain and muscles in your foot. How does the message get from the first neuron to the second?

Once the action potential reaches the end of the first neuron

- the action potential jumps across the synapse and travels down the second neuron.
- a chemical is released that binds to the second neuron and generates an action potential there.
- an electrical signal is sent across the synapse to the second neuron and generates an action potential.
- it synapses with the second neuron to generate an action potential.

16. Your eyes are closed. For one second a person **firmly pushes your left** hand with a stick but just **lightly touches your right** hand with an identical stick. Compare the neural information coming from your left and right hands.

- Action potentials travel faster from your left hand.
- There are more action potentials coming from your left hand.
- The peak of action potentials from your left hand will be higher.
- Each action potential from your left hand will be stronger.

17. A dry leaf falls on your shoulder but you don't feel it. This means the sensory neuron in your shoulder did not reach threshold because

- not enough positive ions moved into the sensory neuron.
- not enough chemical neurotransmitter was released from the sensory neuron.
- the action potentials traveled too slowly down the sensory neuron.
- the peaks of the action potentials on the sensory neuron were too low.